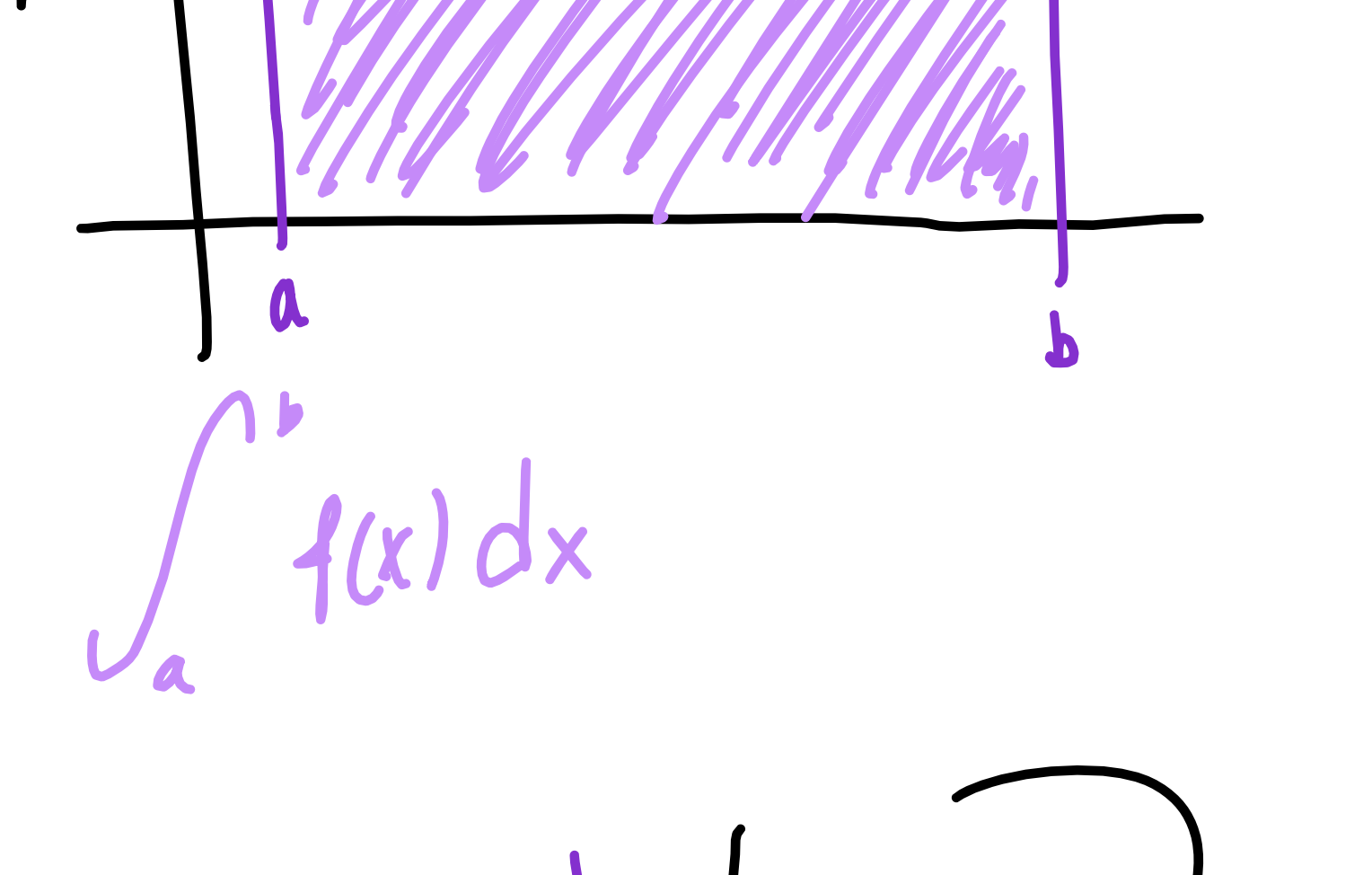


AREA

Section 7.1

1-14-26

A) Area Under a Curve

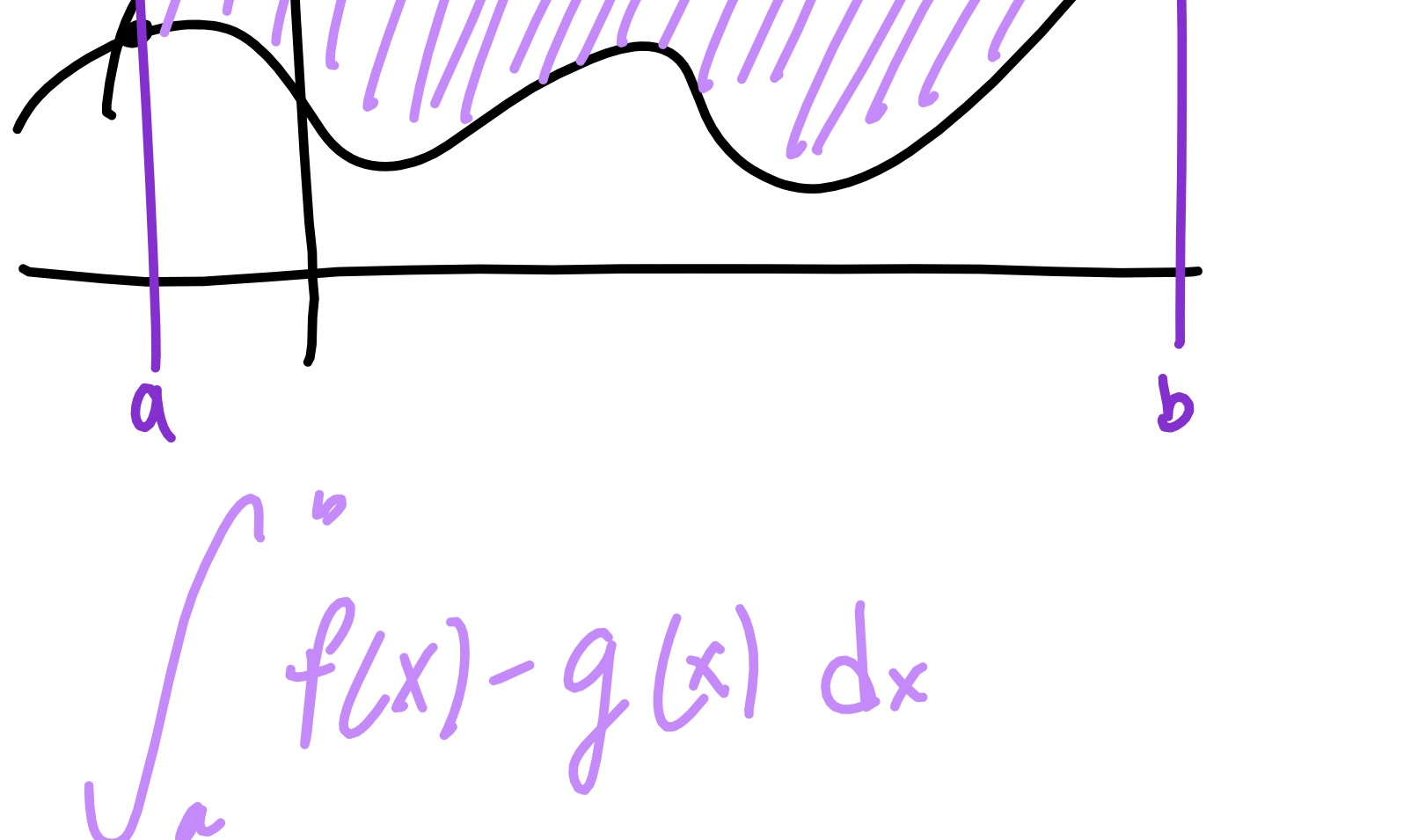


$$\int_a^b f(x) dx$$



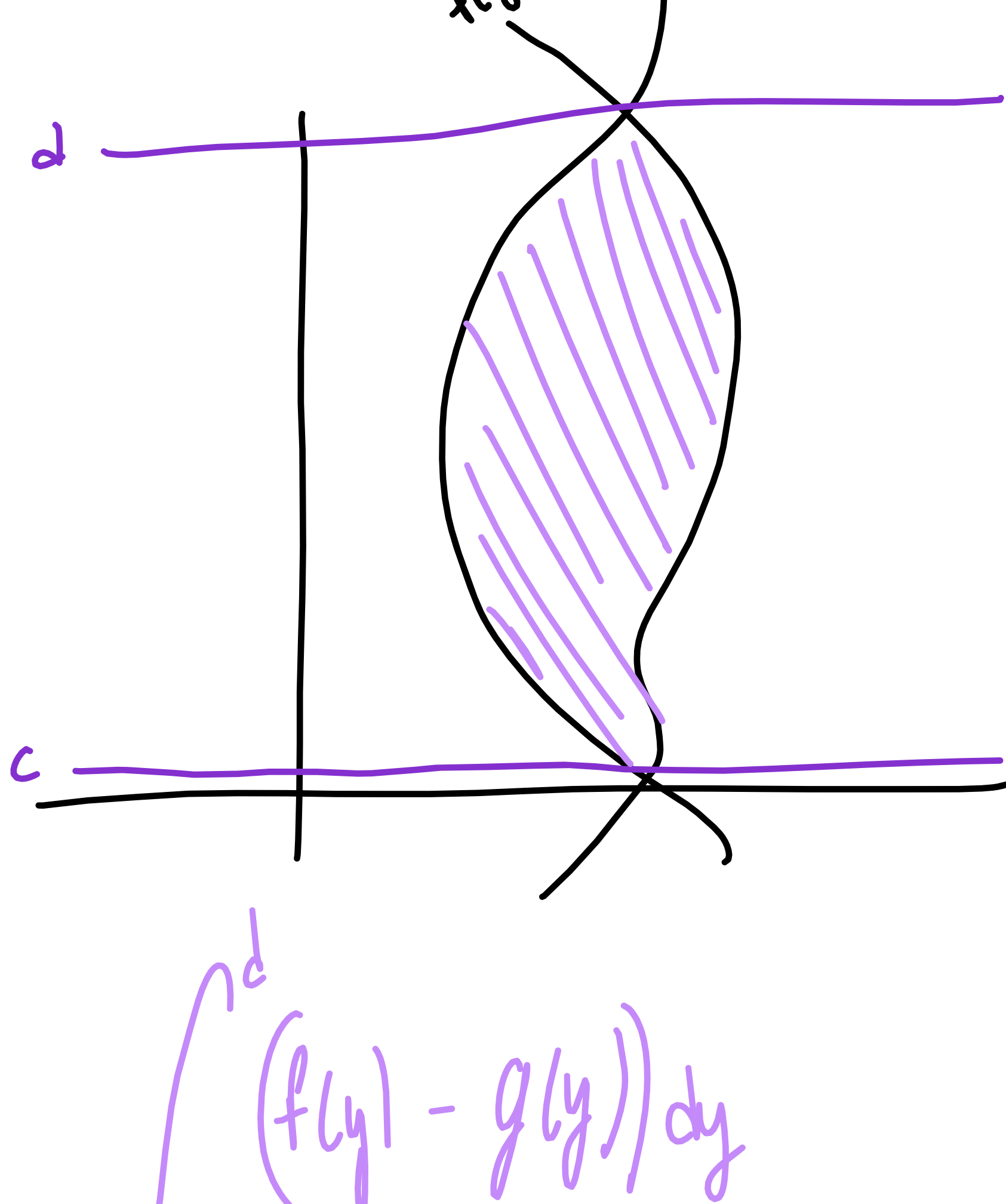
$$\int_c^d f(y) dy$$

B) Area Between 2 Curves



$$\int_a^b f(x) - g(x) dx$$

Top - Bottom



$$\int_c^d (f(y) - g(y)) dy$$

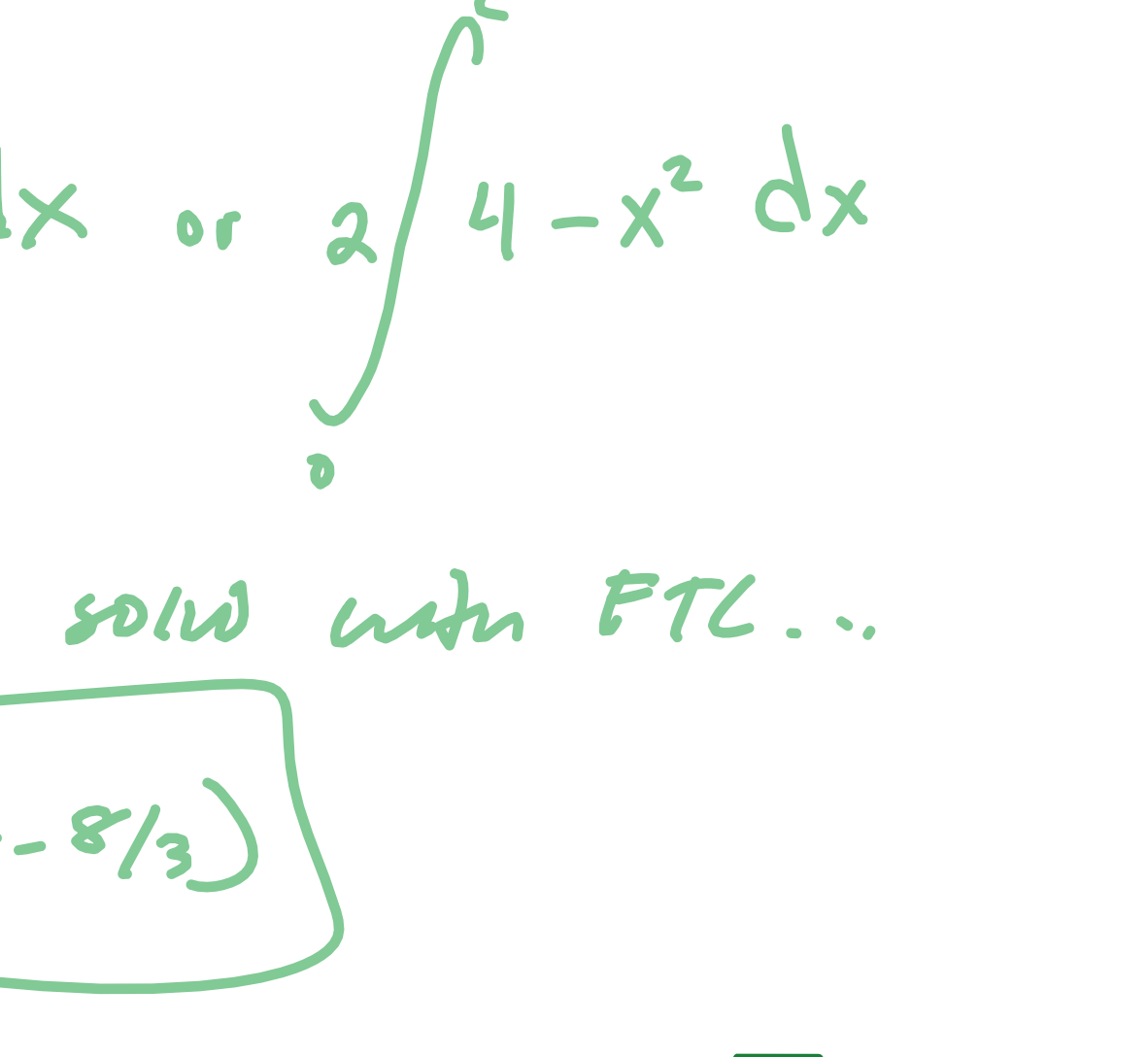
Right - Left

Ex:

$$y = x^2 - 2$$

$$y = 2$$

find endpoints
graphically OR
algebraically



$$\int_{-2}^2 2 - (x^2 - 2) dx$$

$$\int_{-2}^2 4 - x^2 dx \text{ or } 2 \int_0^2 4 - x^2 dx$$

and then solve with FTC...

$$2(8 - 8/3)$$

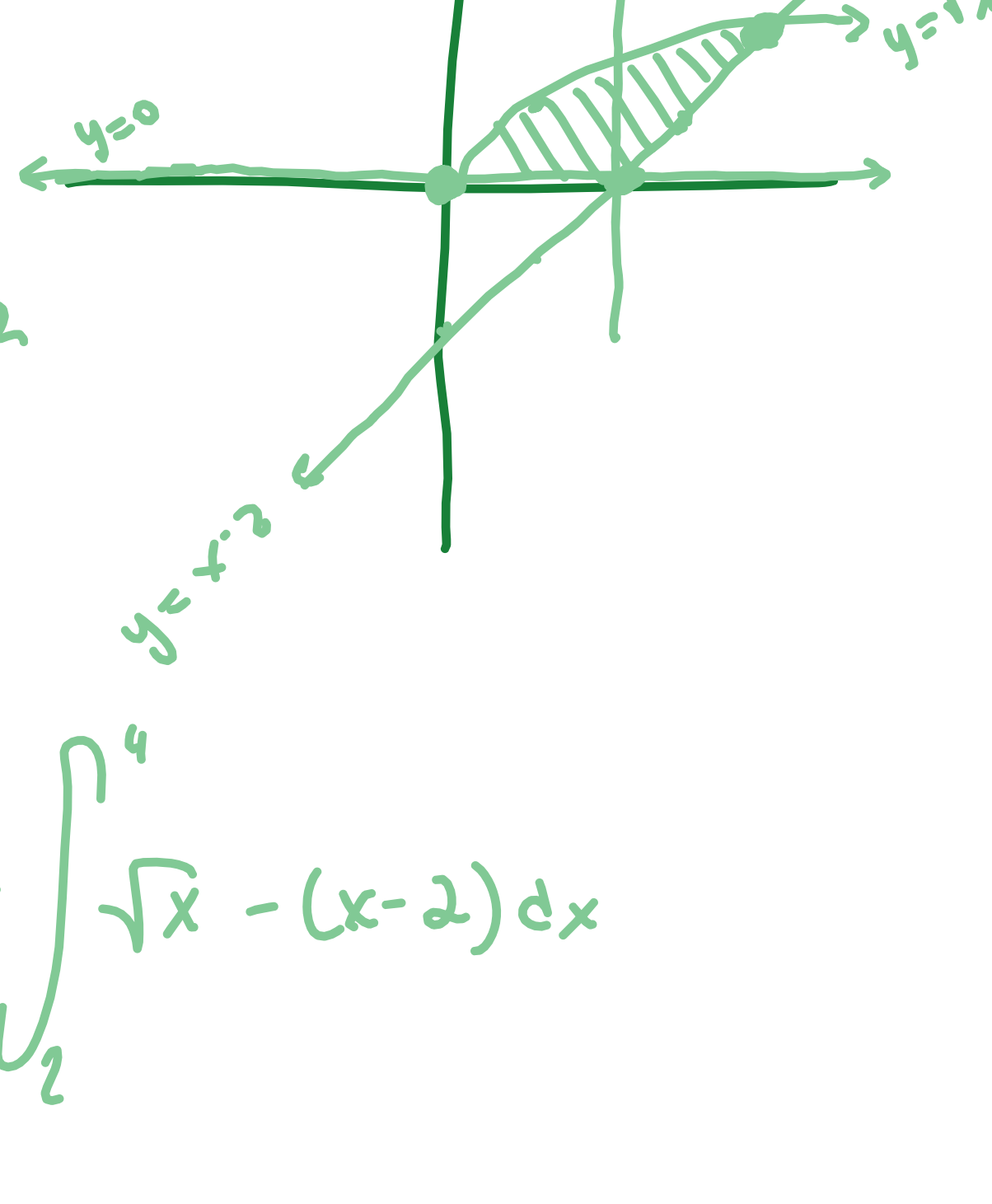
Ex:

$$y = x - 2$$

$$y = \sqrt{x} \text{ or } x = y + 2$$

$$y = 0$$

$$x = y^2$$



$$\int_0^2 \sqrt{x} - 0 dx + \int_2^4 \sqrt{x} - (x - 2) dx$$

$$\int_0^2 y + 2 - y^2 dy$$

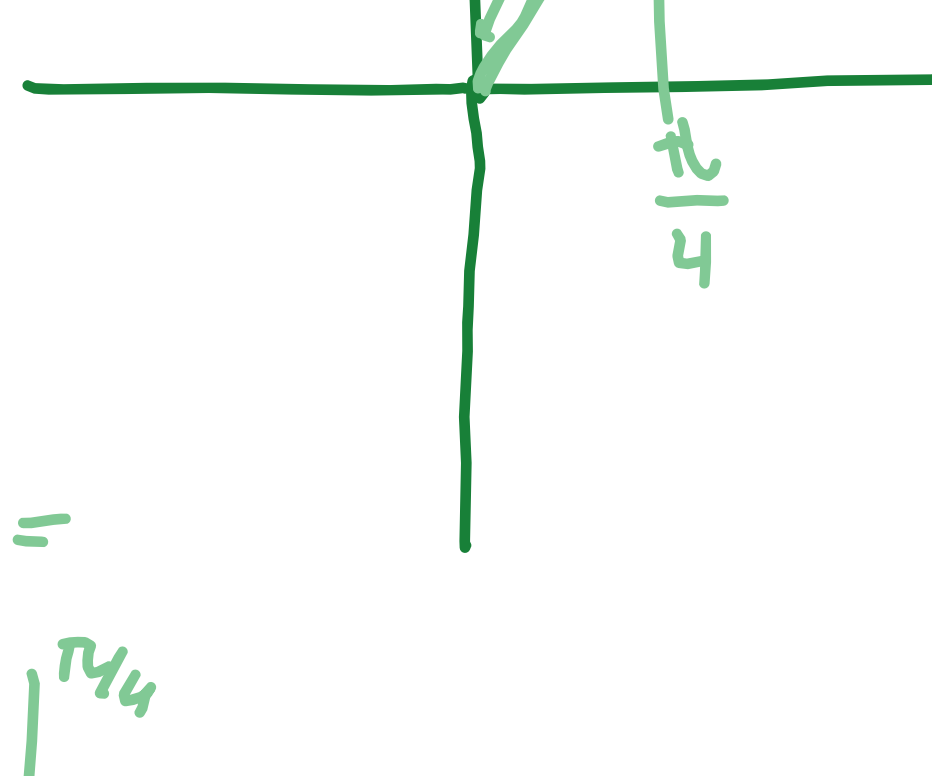
Then solve with FTC

$$\frac{10}{3}$$

Ex:

$$y = \sec^2 x$$

$$y = \sin x$$



$$\int_0^{\pi/4} \sec^2 x - \sin x dx =$$

$$= \tan x + \cos x \Big|_0^{\pi/4}$$

$$= \left(1 + \frac{\sqrt{2}}{2}\right) - (1)$$

$$= \frac{\sqrt{2}}{2}$$

Ex:

$$y = x^3$$

$$x = y^2 - 2$$

$$x = \sqrt[3]{y}$$

$$\text{or } y = \pm \sqrt{x+2}$$



$$\int_{-1}^b \sqrt[3]{y} - (y^2 - 2) dy$$

or

$$\int_{-2}^{-1} (\sqrt{x+2} + \sqrt{x+2}) dx$$